

Simplex method

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Slack variable

A non-negative variable which is added to the left hand side of the constraint to convert $\leq$ to an equation is called a slack variable.

Example

$$3x_1 + x_2 \leq 2$$

$$x_1 + x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

Ans:

$$3x_1 + x_2 + s_1 = 2$$

$$x_1 + x_2 + s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Surplus variable

A non-negative variable which is subtracted from the left hand side of the constraint to convert $\geq$ to an equation is called a slack variable.

Example

$$3x_1 + x_2 \geq 2$$

$$x_1 + x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

Ans:

$$3x_1 + x_2 - s_1 = 2$$

$$x_1 + x_2 - s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Standard form of LPP for applying simplex method

1. All the constraints should be converted to equations except for the non-negativity constraint which remains as inequalities.

Example

$$3x_1 + x_2 \leq 2$$

$$x_1 + x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

Ans:

$$3x_1 + x_2 + s_1 = 2$$

$$x_1 + x_2 - s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

2. Right side element of each constraint should be made non-negative, if not, it can be made positive on multiplying both side of the resulting equation by (-1).

Example

$$x_1 + x_2 \geq -5$$

$$x_1, x_2 \geq 0$$

Ans:

$$x_1 + x_2 - s_2 = -5$$

$$\implies -x_1 - x_2 + s_2 = 5$$

$$x_1, x_2, s_1 \geq 0$$

3. The objective function should be of maximization form. The minimization of a function $f(x)$ is equivalent to the maximization of the negative expression of the function $f(x)$.

Example

$$\text{Min } Z = 3x_1 + x_2$$

Ans: Max $Z' = -3x_1 - x_2$, where $Z' = -Z$

Example

1. Using simplex method, Maximize

$$Z = 3x_1 + 5x_2 + 4x_3$$

subject to

$$2x_1 + 3x_2 \leq 8$$

$$2x_2 + 5x_3 \leq 10$$

$$3x_1 + 2x_2 + 4x_3 \leq 15$$

$$x_1, x_2 \geq 0$$

Step 1: Introduce slack variables

Convert inequalities to equalities by adding slack variables $s_1, s_2, s_3 \geq 0$:

$$2x_1 + 3x_2 + s_1 = 8$$

$$2x_2 + 5x_3 + s_2 = 10$$

$$3x_1 + 2x_2 + 4x_3 + s_3 = 15$$

Initial basic variables: s_1, s_2, s_3 .

Iteration 2: Enter x_3

Most negative reduced cost is -4 (for x_3). Ratios (rows with positive x_3 coeff):

- x_2 : coefficient 0 (skip)
- s_2 : $(14/3)/5 = 14/15 \approx 0.9333$ (smallest)
- s_3 : $(29/3)/4 = 29/12 \approx 2.4167$

Leave variable: s_2 . Pivot on the 5 in row s_2 , column x_3 .

Iteration 2: Tableau after pivot (make x_3 basic)

Divide row s_2 by 5 and eliminate x_3 from other rows. New tableau:

BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS
x_2	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0	$\frac{8}{3}$
x_3	$-\frac{4}{15}$	0	1	$-\frac{2}{15}$	$\frac{1}{5}$	0	$\frac{14}{15}$
s_3	$\frac{41}{15}$	0	0	$-\frac{2}{15}$	$-\frac{4}{5}$	1	$\frac{89}{15}$
Z	$-\frac{11}{15}$	0	0	$\frac{17}{15}$	$\frac{4}{5}$	0	$\frac{256}{15}$

Iteration 3: Enter x_1

Most negative reduced cost: $-11/15$ (for x_1). Ratios for rows with positive x_1 :

- x_2 : $(8/3)/(2/3) = 4$
- x_3 : coefficient $-4/15$ (skip)
- s_3 : $(89/15)/(41/15) = 89/41 \approx 2.1707$ (smallest)

Leave variable: s_3 . Pivot on $41/15$ in the s_3 row, x_1 column.

Iteration 3: Tableau after pivot (make x_1 basic)

After pivoting to bring x_1 into the basis we obtain the final tableau:

BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS
x_2	0	1	0	$\frac{15}{41}$	$\frac{8}{41}$	$-\frac{10}{41}$	$\frac{50}{41}$
x_3	0	0	1	$-\frac{6}{41}$	$\frac{5}{41}$	$\frac{4}{41}$	$\frac{62}{41}$
x_1	1	0	0	$-\frac{2}{41}$	$-\frac{12}{41}$	$\frac{15}{41}$	$\frac{89}{41}$
Z	0	0	0	$\frac{45}{41}$	$\frac{24}{41}$	$\frac{11}{41}$	$\frac{765}{41}$

All reduced costs are non-negative, so the current solution is optimal.

Optimal Solution and Verification

Basic variables from final tableau:

$$x_1 = \frac{89}{41}, \quad x_2 = \frac{50}{41}, \quad x_3 = \frac{62}{41}.$$

Objective value:

$$Z = \frac{765}{41} \approx 18.6585.$$

Each constraint is tight (binding):

$$2x_1 + 3x_2 = 8,$$

$$2x_2 + 5x_3 = 10,$$

$$3x_1 + 2x_2 + 4x_3 = 15.$$

Summary

- Performed three simplex pivots: x_2 entered, then x_3 , then x_1 .
- Final basic feasible solution: $(x_1, x_2, x_3) = (89/41, 50/41, 62/41)$ with $Z = 765/41$.
- All original constraints are binding at the optimum.

Simplex vs Two-Phase Method

Aspect	Simplex Method	Two-Phase Method
Starting BFS	Directly available	Not available
Artificial variables	Not needed	Required
Phases	Single	Two (feasibility + optimality)
Applicability	Limited cases	General, works for all LPPs
Computation	Faster	Longer

Example

1. Maximize

$$Z = 3x_1 + 5x_2 + 4x_3$$

subject to

$$2x_1 + 3x_2 \leq 8, \quad 2x_2 + 5x_3 \leq 10, \quad 3x_1 + 2x_2 + 4x_3 \leq 15, \quad x_1, x_2 \geq 0$$

Maximize

$$Z = 3x_1 + 2x_2$$

subject to

$$x_1 + x_2 \leq 4, \quad 2x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0$$

Solution:

- Add slack variables s_1, s_2 .
- Initial BFS: $(x_1, x_2, s_1, s_2) = (0, 0, 4, 5)$.
- Apply simplex iterations $\Rightarrow (x_1, x_2) = (1, 3), Z_{\max} = 9$.

Two-Phase Method

When to use:

- When initial BFS is not obvious (constraints with $\geq$ or $=$).

Steps:

- 1 **Phase I:** Add artificial variables, minimize sum of artificial variables W .
 - If $W^* > 0$: infeasible problem.
 - If $W^* = 0$: feasible solution obtained.
- 2 **Phase II:** Remove artificial variables and solve original problem using simplex.

Example: Two-Phase Method

Minimize

$$Z = x_1 + x_2$$

subject to

$$x_1 + x_2 \geq 2, \quad x_1, x_2 \geq 0$$

Solution:

- Convert: $x_1 + x_2 - s_1 + a_1 = 2$.
- Phase I: Minimize $W = a_1$.
- Solve Phase I $\Rightarrow W^* = 0$ (feasible).
- Phase II: Solve original $Z = x_1 + x_2 \Rightarrow (x_1, x_2) = (2, 0), Z_{\min} = 2$.